

Realized Persistence Cannot Be Physically Null

Minimal Non-Vacuous Structure for Physical Realization

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Abstract

This paper fixes the first physics-facing threshold in the realized-physics sequence: a regime that carries realized persistence cannot be physically null. Within a fixed admissible continuation regime, a target already typed as standing-bearing persistence cannot collapse to reconstruction-nullity with respect to its own persistence-sensitive continuation structure.

The theorem claim is deliberately minimal. Minimal realization is defined only by stable admissibility, lawful same-domain continuation, a persistence-sensitive reconstruction interface, and fail-closed treatment of standing failure. It does not presuppose topology, metricity, locality, fields, particles, information primitives, observer access, material substrate, or physical occupancy. Given realized persistence, however, admissibility-preserving reconstruction classes cannot all collapse to one null class. A non-trivial continuation-sensitive persistence interior is therefore required.

The physical-role claim is the point of the paper: physical realization first enters as non-null realized persistence, not as spacetime, matter, field content, or dynamics. Later papers strengthen this floor into locality, local-domain organization, metric role, bounded propagation, and field-carrier structure. Those later structures are not absent from the program; they are downstream specializations of the non-vacuous physical-realization role derived here.

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1 Introduction

1.1 Theorem claim and physical role

This article states the first physical-role theorem in the sequence. Its exact theorem claim is reconstruction non-vacuity:

$$\text{MinReal}_D \wedge \text{RealPers}_D \Rightarrow \text{NonVacRec}_D \Rightarrow [\text{Int}_D]_{\equiv_{\text{Int}}} \neq \emptyset.$$

The physical-role claim is the corresponding interpretation: once a fixed admissible regime is required to carry realized persistence, physical realization cannot be null. Standard physical structures are not assumed at this layer, but neither are they merely excluded. They are later specializations of the non-vacuous realized-persistence floor.

Stage	Structural theorem	Physical role
Paper 1	Non-vacuous realized persistence	Physical realization cannot be null
Paper 2	Bounded standing-localization	Structural origin of locality
Paper 3	Continuation-neighborhood domains	Structural origin of local-domain organization
Paper 4	Quotient-stable comparison carrier	Structural origin of the metric role
Paper 5	Bounded perturbation transport	Structural origin of finite-propagation constraint

Thus the non-claim is not “this is not physics.” The correct reading is: this paper derives the first role that any realized physical regime must satisfy before familiar physical specializations can be licensed.

The paper begins with a limited question. Suppose one is already working inside a fixed domain in which admissibility, standing-bearing continuation, lawful redescription, transport discipline, quotient stability, and fail-closed treatment of standing failure are operative. Suppose further that the target is not merely to classify descriptions but to carry persistence: continuation with standing-bearing consequence retained under lawful same-domain continuation. Can such a regime remain null with respect to its own continuation-sensitive reconstruction structure?

The answer proved here is no. A null realization is not a minimal realized persistence regime. If it is completely mute, it remains AMetric pre-admissibility and performs no same-domain work. If it performs admission, rejection, continuation, identity preservation, transport, quotienting, reconstruction, role anchoring, or repair handling, then it has non-vacuous constraint content and is not null. If it supplies an unannounced representative, selector, default, lower generator, or hidden reservoir accessed by the target, that element is non-null governance. If it presents only syntax, trace, semantic reference, or skin-level notation, it does not carry realized persistence. If it changes the target class, it no longer gives a same-domain null realization of the original target.

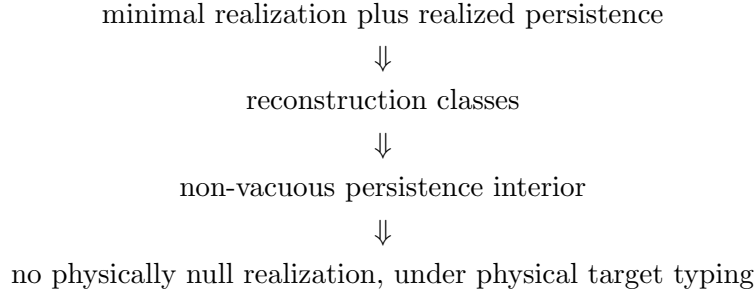
A possible circular reading must be blocked: perhaps “realization” has already been defined as differentiated physical structure, making the theorem circular. The definitions below block that reading by weakening the starting point. Minimal realization is not assumed to be physical, local, metric, topological, field-like, particle-like, information-theoretic, concrete, or materially instantiated. It is only stable admissible continuation capable of persistence-sensitive reconstruction. Differentiation is then derived as a reconstruction consequence. Physical non-vacuity is introduced only after that derivation has been made.

This paper is not a cosmological beginning theorem, a proof that a universe must exist, a derivation of creation from nothing, or a theorem of total ontology. Its target is narrower and more useful for physics-facing reconstruction. It proves that, within the AASC comparison class, realized admissible persistence requires a non-vacuous interior of continuation-sensitive reconstruction. Subsequent work may then ask under which additional persistence constraints that interior becomes standing localization, continuation-neighborhood structure, metric reconstruction,

finite propagation, or field-like carrier structure. None of those subsequent structures is assumed here.

1.2 Core argument

The proof follows a minimal sequence:



The central theorem is not initially a theorem about physics. It is a theorem about reconstruction non-vacuity. Physics enters only when the realized persistence target is required to carry non-dischargeable domain participation rather than remain a bookkeeping description. This delayed typing matters: it prevents the word “physical” from doing any hidden work in the proof.

1.3 What the paper proves

The paper proves four claims in increasing target strength.

- (i) **Minimal realization does not by itself assume differentiation.** A regime may be typed as minimally realized if it has a stable admissibility gate, lawful continuation, persistence-sensitive reconstruction capacity, and fail-closed treatment of standing failure. None of these conditions stipulates locality, topology, metricity, field structure, information structure, material substrate, or physical occupancy.
- (ii) **Realized persistence forces reconstruction non-vacuity.** If minimal realization is asked to carry standing-bearing persistence, the reconstruction classes induced by the regime cannot all collapse to a single null class.
- (iii) **Reconstruction non-vacuity induces a persistence interior.** Non-identical continuation-sensitive reconstruction classes define a nonempty persistence-interior equivalence class under lawful same-domain redescription.
- (iv) **Physical nullity is unavailable as an interpretation of realized persistence.** If the persistence target is physical in the limited sense of non-dischargeable domain participation, the non-vacuous persistence interior excludes physically null realization.

In symbols:

$$\text{MinReal}_D \wedge \text{RealPers}_D \Rightarrow \text{NonVacRec}_D \Rightarrow [\text{Int}_D]_{\equiv_{\text{Int}}} \neq \emptyset,$$

and, with the additional physical target typing,

$$\text{PhysReal}_D \Rightarrow \neg \text{PhysNull}_D.$$

1.4 Claim levels and downstream specializations

The paper has two claim levels. The theorem level proves that realized persistence cannot be reconstruction-null inside the fixed admissible regime. The physical-role level says what that theorem means when the target is physical realization: a physically realized persistence regime must possess non-vacuous continuation-sensitive interior structure.

This does not yet specialize the interior into a universe, metric signature, field content, particle spectrum, coupling constant, Einstein dynamics, quantum dynamics, dark-sector structure, or cosmological initial condition. Those are stronger structures. The present result supplies their first necessary floor. A later physical theory may identify, refine, or represent the persistence interior by spacetime, matter, fields, records, events, or dynamical states, but such identifications require additional role-fixing theorems rather than primitive import.

All uses of “forced”, “necessary”, and “cannot” remain conditional structural claims internal to a fixed admissible continuation regime once realized persistence is in force. They are not primitive metaphysical-necessity claims, cosmogenic claims, or possible-world ensemble claims.

1.5 Category discipline and dependency order

The proof keeps five layers separate. The following table fixes the roles of admissibility, realization, reconstruction, persistence, and differentiation.

Layer	Role in the proof	Not used as
Admissibility	Lawful same-domain admission, rejection, continuation, and failure treatment.	Physical substance, space-time, metricity, probability, or modality.
Minimal realization	A stable setting in which admissible continuation can be carried and reconstructed.	Differentiation, locality, material occupancy, or field structure.
Reconstruction	Domain-internal recovery of continuation-sensitive standing distinctions.	Observer knowledge, coding, information flow, or semantic representation.
Realized persistence	Standing-bearing consequence actually retained under lawful same-domain continuation.	Mere persistence-capability, mere trace, or a possible-world claim.
Differentiation	Non-identical reconstruction classes with standing-relevant contrast.	Primitive topology, object plurality, manifold points, or particles.

The analysis also does not assume a primitive modal ontology, possible-world ensemble, or ensemble semantics. Its claims are conditional structural constraints on a fixed admissible continuation regime once realized persistence is in force.

1.6 Terminology translation table

The manuscript uses AASC terminology because the proof is internal to that framework. The following table gives the neutral structural role of each recurring term. It is not an additional assumption layer.

Framework term	Neutral structural reading	Function in the proof
Admissibility	Lawful same-domain permission or exclusion.	Determines whether a candidate can participate in the evaluated regime.
Standing	Persistence-bearing admitted status under licensed continuation.	Marks retained consequence rather than one-time admission.
Continuation	Admissibility-preserving succession, composition, or carry-forward inside the fixed target.	Supplies the domain in which persistence can be retained or lost.
Realized regime	A stable continuation setting in which persistence-sensitive reconstruction is available.	Blocks purely descriptive or skin-level readings.
Reconstruction	Internal recovery of standing-relevant sameness and difference.	Determines whether persistence is non-vacuous.
Interior	Nonempty quotient class of standing-relevant continuation differentiations.	Names the minimal non-vacuous structure derived by the theorem.
AMetric boundary	Pre-admissible non-selection condition.	Blocks null defaults, hidden selectors, and lower generators.
Skin	Presentation with no standing-bearing role load.	Excludes notation, trace, or description from counting as realization.

1.7 Background within the AASC framework

The result is new in placement rather than in primitives. The AASC corpus already develops the AMetric boundary, bivalence of admissibility, standing as reuse-stable admissibility, fixed-domain exhaustion, null-regime inadmissibility, constraint-bundle forcing, continuation-locus induction, boundary-trace fixation, persistence-regime realization, standing localization, and concrete-domain closure. The present analysis isolates the first structural threshold: non-vacuity of a realized persistence interior.

The important shift is not from abstract admissibility to a specific physics model. It is from admissibility as classification to admissibility as realized persistence. Once persistence is realized, reconstruction-nullity no longer survives.

2 Claim typing and minimal target class

A recurring failure mode in foundational work is type drift. A result about admissible continuation is read as cosmology; a result about non-vacuous domain participation is read as material ontology; a result about AMetric non-selection is read as denial of metrics inside admitted regimes. This manuscript therefore begins by locking the claim type.

Definition 2.1 (Fixed domain). A fixed domain D is a comparison setting in which candidate identity, admissibility status, standing-bearing continuation, lawful redescription, transport behavior, quotient identity, reconstruction behavior, and failure handling are evaluated against a

stable internal target. Changing the admissibility gate, target vocabulary, failure semantics, or comparison relation changes the domain.

Definition 2.2 (Evaluated fragment). The evaluated fragment E_D of a fixed domain is the class of candidates on which the domain’s admissibility and standing predicates are applied. Candidates outside E_D are not rejected occupants; they are outside the evaluated fragment.

Definition 2.3 (Admissibility and standing). $\text{Adm}_D(x) = 1$ means that candidate $x \in E_D$ is admitted for the relevant same-domain use. $\text{Stand}_D(x) = 1$ means that x retains reuse-stable admissibility under the licensed continuations and redescrptions of D . Standing denotes the persistence-bearing admissibility status required for continuation-sensitive reconstruction under lawful continuation. It is a functional role: it marks whether consequence can be retained through licensed same-domain continuation without reset, transfer, or post-failure repair. Standing is stronger than one-time admission and weaker than any specific physical ontology.

Definition 2.4 (Persistence-capability). PersCap_D holds when the domain has the type of admissibility gate, continuation vocabulary, failure semantics, and reconstruction interface needed to ask whether standing-bearing consequence can persist. Here continuation means only admissibility-preserving succession or composition internal to the fixed regime; it is not temporal flow, causal propagation, or dynamical evolution unless later structure supplies those interpretations. Persistence-capability is a target capacity. It does not imply that any candidate actually persists.

Definition 2.5 (Minimal realization). MinReal_D holds when D supports stable admissible continuation capable of persistence-sensitive reconstruction. More explicitly, minimal realization requires:

- M1.** a stable admissibility gate on the evaluated fragment;
- M2.** lawful same-domain continuation rather than mere external description;
- M3.** a reconstruction interface able in principle to retain or lose standing-relevant consequence;
- M4.** fail-closed treatment of standing failure, so that failure is not repaired as the same evaluated act.

Minimal realization does not assume interior differentiation, topology, metricity, locality, causal order, field structure, particle ontology, material substrate, modal ontology, information primitives, or physical occupancy. Its reconstruction requirement is only that standing-relevant continuation can be retained or lost by licensed same-domain criteria.

Definition 2.6 (Realized persistence). RealPers_D holds when minimal realization is not merely available but actually carries standing-bearing consequence through lawful same-domain continuation. Thus RealPers_D implies MinReal_D , but MinReal_D alone need not imply RealPers_D .

Remark 2.7 (Why the distinction matters). Persistence-capability is a typed possibility of asking the persistence question. Minimal realization is a stable setting in which the question can be operationally carried within the domain. Realized persistence is the successful carrying of standing-bearing consequence. The main theorem uses the third notion, but its non-circularity depends on the second notion being minimal and non-geometric.

Definition 2.8 (Realized physical target). PhysReal_D holds when RealPers_D is interpreted as non-dischargeable participation in the target domain rather than as description, notation, bookkeeping, or external semantic reference. The word “physical” here does not mean manifold, metric, matter, field, particle, observer, or spacetime. It means that the persistence is borne by the target regime itself.

Definition 2.9 (Physically null realization). PhysNull_D holds when a same-domain candidate is claimed to satisfy PhysReal_D while possessing no continuation-sensitive reconstruction differentiation, no standing-relevant interior role, no admissibility-bearing transport, no role-anchor load, and no non-skin tensor content. It is therefore stronger than emptiness of occupants and stronger than absence of a chosen representative.

Remark 2.10 (No topology or metric is imported). The words “interior”, “locus”, and “neighborhood” in this paper are not topological primitives. They name quotient roles induced by admissibility-bearing continuation and reconstruction. No open sets, manifold charts, metric distances, differentiable structure, or primitive adjacency relation are assumed. Those are subsequent reconstruction targets.

2.1 Governance non-vacuity versus target-domain non-vacuity

There is an important distinction between empty admission and null realization. A domain can have governance without admitted occupants. For example, a gate may reject every candidate in a fragment. Such a gate is not target-domain non-vacuous merely because the gate exists. It is governance-non-vacuous. The present theorem does not infer target-domain interiority from empty admission.

The theorem instead concerns realized persistence. If the target is to carry standing-bearing consequence, then total reconstruction-nullity is incompatible with that target. This avoids the weak argument: “there is a rule, therefore there is a target-domain interior.” The correct argument is: “there is realized standing-bearing continuation, therefore the reconstruction structure carrying that continuation is non-vacuous.”

Proposition 2.11 (No occupant inference from empty admission). *A fixed domain may possess an admissibility gate Adm_D and yet admit no occupants in a chosen fragment. Such emptiness does not imply PhysReal_D , RealPers_D , or NonVacInt_D .*

Proof. A rejecting gate can be governance-non-vacuous while no positive standing-bearing continuation is present. The target of realized persistence requires retained consequence, not merely gate existence. Therefore empty admission supplies no occupant-level target realization. $\square$

Proposition 2.12 (Physical non-vacuity from realized persistence, not from admission). *If non-vacuity is inferred in this paper, it is inferred from RealPers_D plus minimal realization, not from the bare existence of Adm_D .*

Proof. The reconstruction argument requires continuation-sensitive classes and fail-closed standing behavior. Bare admission does not supply those. Hence the non-vacuity theorem is not a governance-to-physics inference. $\square$

3 Preliminaries: the AASC roles used here

The paper uses only a restricted part of the AASC spine. These dependencies are stated as role constraints rather than reproved from first principles.

3.1 The AMetric boundary

The AMetric boundary is not a metric space with zero distance, a null point, a vacuum, or an empty set. It is the pre-admissible condition in which no admissibility-bearing distinction, selector, ordering, representation, quotient, probability, metric, continuation, or gate authority has yet been admitted. Because no selector is available there, AMetric nullity cannot select itself as the default realization.

Dependency 3.1 (AMetric non-selection). At the pre-admissible boundary, no candidate is selected, ranked, metricized, probabilistically weighted, indexed, or canonically represented by default. A purported boundary selector either imports admitted structure, changes scope, collapses to bookkeeping, or remains inert.

3.2 Bivalence and no third status

On the evaluated fragment of a fixed domain, admissibility-relevant status is bivalent: a candidate is admitted for the relevant use or it is not. Same-side distinctions either alter licensed use, in which case they are hidden gate work, or they do no admissibility work and quotient away as skin or bookkeeping.

Dependency 3.2 (No null-but-admissible status). There is no same-scope status under which a candidate is null yet admissibility-bearing, failed yet standing, outside yet governing, skin-only yet authoritative, or AMetric yet selecting. Such statuses either collapse to one side of the gate or change the target class.

3.3 Standing conservation and no same-act repair

Standing can be preserved or destroyed under licensed same-domain continuation. It cannot be created, amplified, transferred, or repaired as the same evaluated act after standing failure. A genuine improvement requires explicit reset and fresh evaluation of a new act or a new scope.

Dependency 3.3 (Standing failure non-repairability). If $\text{Stand}_D(x) = 0$, no internal same-scope post-processing that still treats the result as the same evaluated act can produce $\text{Stand}_D(x) = 1$. Repair either fails, supplies non-null governance, or triggers fresh evaluation.

3.4 Anchor–tensor–skin discipline

AASC role discipline separates admissibility-bearing content into anchors, tensors, and skin. Anchors are preconditions or boundary roles that define admissibility-relevant eligibility. Tensors are invariant standing-bearing content preserved through quotient and redescription. Skin is presentation: notation, coordinate choice, labeling, encoding, syntactic display, or representational surplus with no gate authority.

A null realization has no anchor and no tensor. If it has only skin, it cannot govern or preserve realized persistence. If it has anchor or tensor load, it is not null.

Dependency 3.4 (ATS role exhaustion). Every admissibility-bearing descriptive element in the fixed comparison class is anchor, tensor, or skin. Skin has no independent admissibility-bearing authority.

3.5 Continuation loci and role anchors

Standing-bearing continuation induces a continuation locus: the minimally preserved admissibility-relevant persistence structure under lawful same-domain continuation. For that locus to participate in a fixed role, a role-anchor class must be fixed. Boundary-trace fixation supplies such role-anchor fixation when internal structure is non-selective but an admissible bivalent trace fixes a role quotient without selecting a representative.

The present paper does not require the full boundary-trace theorem. It requires only the consequence that realized persistence is not a raw sequence. It is locus-bearing, role-bearing, quotient-stable, and reconstruction-sensitive.

Dependency 3.5 (Continuation-locus induction). Every non-degenerate continuation-bearing governance regime induces a continuation-locus equivalence class $[L_D]_{\equiv_{\text{Loc}}}$. Standing-retained continuation induces positive-side occupant structure at the quotient level.

Dependency 3.6 (Role-anchor fixation). When a continuation locus participates in a fixed admissibility-bearing role and internal structure does not select a canonical representative, an admissible boundary trace can fix a role-anchor equivalence class $[R_D]_{\equiv_{\text{Role}}}$ without selecting a representative.

3.6 Persistence-regime realization

Persistence-regime realization packages the movement from standing-bearing continuation to consequence-retaining persistence. A realized persistence regime factors through continuation loci, role-anchor fixation, trajectory structure, persistence-bearing consequence retention, and persistence-regime equivalence.

Dependency 3.7 (Persistence-regime realization). If irreversible standing-bearing continuation is retained in a fixed domain, then the domain induces a persistence-regime equivalence class $[P_D]_{\equiv_{\text{Pers}}}$. This equivalence class is stronger than syntactic trace and weaker than any specified physical ontology.

4 Reconstruction-sensitive continuation

The main theorem is easiest to state after defining reconstruction-sensitive continuation. This section is the formal center of the revision. It supplies the missing non-circular bridge from minimal realization to interior differentiation.

Definition 4.1 (Continuation history). A continuation history $h \in \text{Hist}_D$ is a finite or schematically finite same-domain record of admissibility-relevant continuation acts, failures, standing retention, and lawful redescrptions in D . A history is not a time series unless later structure supplies temporal interpretation.

Definition 4.2 (Reconstruction interface). A reconstruction interface Rec_D is the fixed-domain role by which admissibility-relevant histories are tested for standing retention, standing loss, lawful redescription, failure behavior, transport role, quotient identity, and role participation. It is not an observer, measurement device, semantic interpreter, information channel, entropy functional, or coordinate chart. It is the domain-internal criterion by which persistence-sensitive sameness and difference are retained or lost.

Definition 4.3 (Admissible persistence-sensitive reconstruction maps). Let $\mathcal{F}_D$ be the class of admissibility-preserving persistence-sensitive reconstruction maps licensed inside D . Each $r \in \mathcal{F}_D$ has the form

$$r : \text{Hist}_D \longrightarrow O_r,$$

where O_r is an admitted output role for standing retention, failure status, lawful redescription, transport behavior, quotient identity, or role participation. A map belongs to $\mathcal{F}_D$ only if it satisfies the following admissibility constraints:

- R1. Same-domain licensing.** It is licensed by the fixed target D , not by an external semantic reader, observer convention, or later scope change.
- R2. Admissibility preservation.** It does not identify admitted standing-retaining histories with standing-failing histories unless the target has explicitly collapsed that distinction, in which case realized persistence is lost.
- R3. Continuation compatibility.** It respects lawful same-domain continuation and does not manufacture distinctions that no licensed continuation or failure behavior can bear.
- R4. Standing-bearing distinguishability.** It distinguishes only contrasts whose collapse would alter standing retention, standing failure, role participation, transport behavior, quotient identity, or lawful redescription.
- R5. Lawful composability.** If histories compose by licensed continuation, the reconstruction of the composite is compatible with the reconstructions of its licensed parts, up to lawful same-domain redescription.
- R6. Quotient and role invariance.** It is invariant under skin-level presentation changes and respects role-anchor and quotient discipline.
- R7. Fail-closedness.** It cannot repair standing failure as the same evaluated act by representational post-processing.

The maps in $\mathcal{F}_D$ are not measurements by an observer. They are fixed-domain tests of persistence-bearing consequence. These constraints prevent arbitrary reconstruction families from manufacturing the desired differentiation.

Remark 4.4 (What is being reconstructed). Reconstruction concerns continuation-sensitive distinguishability relations internal to the realized persistence regime: which continuations retain standing, which lose it, which redescriptions are lawful, which roles are preserved, and which quotient identities remain stable. It does not reconstruct observer information, semantic content, coordinate descriptions, epistemic accessibility, or external representation.

Lemma 4.5 (No arbitrary-map manufacture). *A purported distinction generated only by a map that fails one of **R1–R7** is not a reconstruction distinction for D .*

Proof. By definition, $\mathcal{F}_D$ contains only maps satisfying the admissibility constraints **R1–R7**. A map that violates same-domain licensing, admissibility preservation, continuation compatibility, standing-bearing distinguishability, lawful composability, quotient/role invariance, or fail-closedness is external bookkeeping, skin, hidden governance, or a scope change. Such a map cannot witness NonVacRec_D . $\square$

Theorem 4.6 (Reconstruction Constraint Necessity). *The restrictions **R1–R7** are not optional strengthening assumptions. They are minimal conditions for persistence-sensitive reconstruction in a fixed domain. If a proposed reconstruction family fails same-domain licensing, admissibility preservation, continuation compatibility, standing-bearing distinguishability, lawful composability, quotient/role invariance, or fail-closedness, then it either manufactures distinctions not borne by the target, erases standing-relevant consequence, repairs failure by representation, imports hidden governance, or changes scope. In none of those cases does it reconstruct realized persistence for D .*

Proof. A persistence-sensitive reconstruction family must track the same evaluated target through licensed continuation. Failure of **R1** makes the map external to the target. Failure of **R2** erases the difference between admitted standing and standing failure. Failure of **R3** disconnects the alleged contrast from lawful continuation. Failure of **R4** permits contrasts whose collapse changes no standing-bearing consequence, hence skin. Failure of **R5** prevents reconstruction of composed continuation from remaining compatible with reconstruction of the licensed parts. Failure of **R6** mistakes presentation or representative choice for role identity. Failure of **R7** permits post-hoc representational repair of a failed act. Each failure destroys, externalizes, or replaces persistence-sensitive reconstruction. Therefore the constraints are forced by the role of reconstruction itself, not chosen to make the theorem work. $\square$

Definition 4.7 (Reconstruction equivalence). For histories $h_1, h_2 \in \text{Hist}_D$, write $h_1 \equiv_{\text{Rec}} h_2$ iff every map in the constrained admissible family $\mathcal{F}_D$ identifies them:

$$h_1 \equiv_{\text{Rec}} h_2 \quad :\Longleftrightarrow \quad \forall r \in \mathcal{F}_D, r(h_1) = r(h_2).$$

Equivalently, no licensed same-domain reconstruction distinguishes them with respect to standing-bearing consequence, failure behavior, transport role, quotient identity, role participation, or lawful redescription. The reconstruction classes are

$$\mathcal{R}_D := \text{Hist}_D / \equiv_{\text{Rec}}.$$

Remark 4.8 (Not an information-theoretic primitive). The word “reconstruction” does not mean observer knowledge, entropy, message content, coding capacity, measurement accessibility, or information processing. Those notions may be admitted in later specialized regimes, but they are not primitive here. The primitive role is admissibility-bearing persistence-sensitive reconstruction: whether a same-domain continuation retains, loses, or lawfully redescibes standing-bearing consequence.

Definition 4.9 (Continuation-sensitive reconstruction distinction). There is a continuation-sensitive reconstruction distinction in D when there exist classes $C_1, C_2 \in \mathcal{R}_D$ with $C_1 \neq C_2$. Such a distinction is standing-relevant only if collapsing C_1 and C_2 would alter retained standing, failure status, role participation, transport behavior, quotient identity, or lawful redescription.

Definition 4.10 (Reconstruction non-vacuity). NonVacRec_D holds iff $\mathcal{R}_D$ contains non-identical standing-relevant continuation-sensitive reconstruction classes:

$$\text{NonVacRec}_D \quad :\Longleftrightarrow \quad \exists C_1, C_2 \in \mathcal{R}_D (C_1 \neq C_2) \text{ with standing-relevant contrast.}$$

Remark 4.11 (Why this does not assume locality). A reconstruction class need not be a point, region, event, state, field value, particle, or spacetime location. It is only a quotient role in the persistence analysis. The later emergence of continuation-neighborhood structure is a separate subsequent reconstruction target.

Remark 4.12 (Not merely Platonic distinction). The non-vacuity theorem does not rely on bare set-theoretic difference between abstract objects. A contrast counts only when it is internally persistence-bearing, continuation-sensitive, and reconstruction-active: collapsing it would alter standing retention, failure behavior, transport role, quotient identity, role participation, or lawful redescription inside the fixed domain. Merely formal distinguishability with no such role is skin or bookkeeping.

Lemma 4.13 (Reconstruction collapse destroys persistence discrimination). *If all histories in Hist_D collapse to a single reconstruction class, then D cannot discriminate standing retention from standing failure, lawful continuation from unlawful replacement, identity-preserving redescription from scope change, or role participation from skin-level presentation.*

Proof. Standing retention, standing failure, lawful redescription, role participation, and scope change are distinct admissibility-relevant roles by the definitions of Stand_D , MinReal_D , and fail-closed standing. If all histories occupy a single reconstruction class, then every map in $\mathcal{F}_D$ identifies those roles and no domain-internal reconstruction can distinguish them. The purported distinctions would then be supplied either by an external semantic reader, by skin, or by a hidden selector. External semantics and skin do not carry standing-bearing consequence; a hidden selector is non-null governance. Therefore total reconstruction collapse destroys persistence discrimination. $\square$

Theorem 4.14 (Reconstruction Non-Vacuity). *For any fixed domain D ,*

$$\text{MinReal}_D \wedge \text{RealPers}_D \implies \text{NonVacRec}_D.$$

Equivalently, a minimally realized regime carrying standing-bearing persistence must induce non-identical continuation-sensitive reconstruction classes.

Proof. Assume $\text{MinReal}_D \wedge \text{RealPers}_D$. Minimal realization supplies lawful continuation, persistence-sensitive reconstruction capacity, and fail-closed standing treatment. Realized persistence requires that standing-bearing consequence is actually retained through some lawful same-domain continuation and that failure is not counted as the same retained standing. Suppose, for contradiction, that $\neg \text{NonVacRec}_D$. Then all histories relevant to standing retention, standing failure, lawful redescription, transport, quotient identity, and role participation are identified by every map in $\mathcal{F}_D$ and collapse to a single reconstruction class. By the previous lemma, the domain cannot discriminate retention from failure or lawful continuation from replacement. That contradicts RealPers_D , since persistence requires retained standing rather than indistinguishable collapse. Therefore NonVacRec_D . $\square$

Corollary 4.15 (Persistence-capability is not enough). *PersCap_D alone does not imply NonVacRec_D . The non-vacuity follows only once persistence is realized inside a minimally realized regime.*

Proof. Persistence-capability permits the question to be asked. It does not require a positive standing-bearing continuation to be carried. The theorem uses RealPers_D , not PersCap_D alone. $\square$

5 Null realization instability

We can now state null instability without importing physicality. The first target is reconstruction-nullity. Physical nullity is a later interpretation.

Definition 5.1 (Reconstruction-null regime). A reconstruction-null regime is a same-domain candidate in which all continuation histories relevant to standing, failure, transport, quotient identity, lawful redescription, and role participation collapse to a single null reconstruction class, with no non-skin anchor or tensor load distinguishing them.

Lemma 5.2 (Null indistinguishability collapse). *A reconstruction-null regime has no internal basis for distinguishing admission from rejection, standing from failure, lawful continuation from replacement, same-domain redescription from scope change, or role participation from skin-level presentation.*

Proof. By definition, reconstruction-nullity removes all standing-relevant reconstruction distinctions. The listed contrasts are standing-relevant by the role definitions of the target class. Therefore no internal basis remains. $\square$

Lemma 5.3 (Persistence failure under null collapse). *If a minimally realized regime becomes reconstruction-null while still claiming to carry realized persistence, then it loses the distinction required to treat persistence as persistence rather than failure, replacement, or skin-level record.*

Proof. Realized persistence requires the retained standing of a continuation-bearing candidate. If retention and failure are reconstruction-equivalent, no same-domain basis remains for assigning standing retention rather than failure. Any later assignment would be a hidden selector, a new scope, or external bookkeeping. None is the original same-domain realized persistence. $\square$

Proposition 5.4 (Null realization instability). *A reconstruction-null regime cannot preserve realized standing-bearing persistence:*

$$\text{RealPers}_D \Rightarrow \neg \text{ReconstructionNull}_D.$$

Proof. This is the contrapositive of the Reconstruction Non-Vacuity Theorem restricted to reconstruction-nullity. If all reconstruction classes collapse, NonVacRec_D fails; but $\text{MinReal}_D \wedge \text{RealPers}_D$ implies NonVacRec_D . Hence reconstruction-nullity is incompatible with realized persistence. $\square$

5.1 Why this is not a performative contradiction

The argument does not say that describing nullity makes nullity non-null. A description can be skin-level notation. The claim is narrower: if nullity is made to carry admissibility-bearing continuation, standing retention, role fixation, transport, reconstruction, or fail-closed consequence, then it is no longer null in the target sense. If it is not made to carry those roles, it does not realize persistence.

5.2 Realization boundary collapse

A common escape route says that differentiation may vanish while realization remains latent, primitive, or lower-level. The following theorem is the explicit boundary result needed to close that route.

Theorem 5.5 (Realization Boundary Collapse). *If a candidate regime in D loses all continuation-sensitive reconstruction differentiation while claiming to preserve $\text{MinReal}_D \wedge \text{RealPers}_D$, then exactly one of the following occurs:*

- (a) it exits minimal realization and becomes *AMetric*, *inert*, or *unevaluated*;
- (b) it retains only *skin*, *trace*, or *external semantic description* and therefore does not carry realized persistence;
- (c) it imports a non-null selector, witness, gate, quotient, role anchor, transport rule, access relation, or memory;
- (d) it changes scope and is no longer the same-domain candidate.

In no case is there a same-domain reconstruction-null realized persistence regime.

Proof. If all continuation-sensitive reconstruction differentiation is lost, then the regime cannot internally distinguish standing retention from failure, lawful continuation from replacement, or redescription from scope change. If no additional role is supplied, it fails the reconstruction and fail-closed components of MinReal_D and exits the target class, giving (a). If only syntax, trace, or semantic reference remains, skin carries no standing-bearing authority by ATS role exhaustion, giving (b). If a distinction is nevertheless recovered, it must be supplied by a selector, witness, gate, quotient, role anchor, transport rule, access relation, or memory; each is non-null governance, giving (c). If the recovery changes the admissibility gate, evaluated fragment, failure semantics, or target vocabulary, the fixed domain has changed, giving (d). These options exhaust the same-domain boundary cases, so none yields reconstruction-null realized persistence. $\square$

Corollary 5.6 (Hidden Reconstruction Reservoir Exclusion). *Realized persistence requires admissible continuation-sensitive accessibility internal to the regime itself. A latent reservoir of distinctions that is inaccessible to all admissibility-preserving persistence-sensitive reconstruction maps in D is realization-inert for D and cannot function as realized persistence structure. If the reservoir becomes accessible to such maps, then the access relation is itself non-null governance or an explicit scope change.*

Proof. If the alleged reservoir is inaccessible to every $r \in \mathcal{F}_D$, then it makes no difference to standing retention, failure status, transport behavior, quotient identity, role participation, or lawful redescription in the fixed domain. It therefore cannot carry realized persistence for D . This is not an epistemic condition: the issue is not whether an observer can know the reservoir, but whether admissible continuation-sensitive access is available inside the target regime. If the reservoir does make such a difference, then some reconstruction map, selector, witness, trace memory, role anchor, access relation, or transport rule connects it to D ; that connector is non-null governance unless the target domain has changed. Hence hidden inaccessible differentiation does not rescue null collapse. $\square$

Corollary 5.7 (No lower physical null generator). *A lower or latent layer cannot generate realized persistence while remaining null. It either remains *AMetric* and *inert*, becomes *skin-only description*, imports non-null governance, or changes the domain.*

Proof. Apply the Realization Boundary Collapse Theorem and Hidden Reconstruction Reservoir Exclusion to the alleged lower layer. Generation is a continuation-sensitive role. If it is absent or inaccessible to $\mathcal{F}_D$, the layer is inert. If present, the layer is non-null governance or a different domain. $\square$

Corollary 5.8 (No default realized target). *Nullity cannot be selected as the default realized persistence regime merely by lacking competitors.*

Proof. Default selection is a selector. At the AMetric boundary no selector is available. Inside an admitted regime a selector is non-null governance. Therefore default null realization is unavailable. $\square$

5.3 The role contradiction

The nullity/realization incompatibility can be summarized as a role contradiction. Realized persistence requires at least the following roles:

- (i) a standing-positive continuation distinction;
- (ii) a fail-closed distinction between retention and loss;
- (iii) a quotient-stable identity condition through lawful redescription;
- (iv) a reconstruction interface that can preserve or lose consequence;
- (v) a boundary or role-anchor discipline preventing hidden representative selection.

Reconstruction-nullity denies all of them. Therefore nullity cannot perform the role of realized persistence while remaining null.

6 Interior differentiation necessity

Reconstruction non-vacuity gives the minimal differentiated content. The paper calls its equivalence class the persistence interior. This is a role-theoretic term, not a geometric one.

Definition 6.1 (Interior differentiation). A domain D has interior differentiation when there exist non-identical continuation-sensitive reconstruction classes whose contrast is standing-relevant:

$$\text{Diff}_D := \{(C_1, C_2) \in \mathcal{R}_D^2 : C_1 \neq C_2 \text{ and the contrast affects standing-relevant reconstruction}\}.$$

Thus $\text{Diff}_D \neq \emptyset$ iff NonVacRec_D .

Lemma 6.2 (Standing-bearing continuation induces differentiation). *If a domain carries standing-bearing continuation, then it must distinguish at least standing-retaining continuation from standing-losing failure.*

Proof. Standing-bearing continuation is not merely succession. It is continuation under which standing is retained. Fail-closed standing requires that standing loss is not treated as the same retained act. Hence at least one standing-relevant reconstruction contrast is required. $\square$

Lemma 6.3 (Realized persistence strengthens differentiation). *If continuation is realized as persistence, the required distinction must be stable under lawful same-domain redescription.*

Proof. If the distinction survives only a notation or presentation and not lawful redescription, it is skin. Skin cannot carry standing-bearing consequence. Realized persistence therefore requires quotient-stable reconstruction contrast. $\square$

Proposition 6.4 (Persistence Differentiation).

$$\text{MinReal}_D \wedge \text{RealPers}_D \implies \text{Diff}_D \neq \emptyset.$$

Proof. By Reconstruction Non-Vacuity, $\mathcal{R}_D$ contains non-identical standing-relevant reconstruction classes. Those classes define an element of Diff_D . Hence $\text{Diff}_D \neq \emptyset$. $\square$

6.1 Minimal persistence interior

Definition 6.5 (Persistence-interior equivalence). Two standing-relevant differentiation structures $\Delta_1, \Delta_2 \in \text{Diff}_D$ are persistence-interior equivalent, written $\Delta_1 \equiv_{\text{Int}} \Delta_2$, when lawful same-domain redescription preserves their standing-retention, failure, transport, quotient, and role-anchor behavior.

Definition 6.6 (Minimal persistence interior). The minimal persistence interior of D is the equivalence class

$$[\text{Int}_D]_{\equiv_{\text{Int}}} := \text{Diff}_D / \equiv_{\text{Int}}$$

when $\text{Diff}_D \neq \emptyset$. It is an interior because it supplies the internal standing-relevant differentiation required for persistence. It is minimal because it does not select a representative, geometry, field, particle, or substance.

Proposition 6.7 (Minimal Persistence Interior). *For any fixed domain D ,*

$$\text{MinReal}_D \wedge \text{RealPers}_D \implies [\text{Int}_D]_{\equiv_{\text{Int}}} \neq \emptyset.$$

Proof. By Persistence Differentiation, $\text{Diff}_D \neq \emptyset$. Quotienting a nonempty class by lawful same-domain redescription yields a nonempty equivalence class. Thus $[\text{Int}_D]_{\equiv_{\text{Int}}} \neq \emptyset$. $\square$

Corollary 6.8 (No representative realism). *The theorem does not select a unique representative of the persistence interior. It proves only that the role-equivalence class is nonempty.*

Proof. The construction quotients by $\equiv_{\text{Int}}$. Selection of a representative would require an additional selector. No such selector is introduced. $\square$

6.2 Interior is not multiplicity

The result does not require a large number of objects, points, states, particles, or possible worlds. The minimal distinction is role-theoretic: retained standing must be distinguished from standing failure or replacement.

Proposition 6.9 (Multiplicity irrelevance). *Numerical multiplicity of representatives is neither necessary nor sufficient for the persistence interior. What matters is standing-relevant reconstruction contrast under lawful continuation.*

Proof. Many representatives that quotient to the same standing-relevant class add no interior differentiation. Conversely, a minimal quotient contrast between retained standing and failure is already non-vacuous. Therefore multiplicity of representatives is not the target criterion. $\square$

6.3 Interior is not metricity

Proposition 6.10 (Pre-metric status of the interior). *The existence of $[\text{Int}_D]_{\equiv_{\text{Int}}} \neq \emptyset$ does not imply a metric, topology, manifold, causal cone, measure, probability space, or coordinate chart.*

Proof. The interior is defined by reconstruction classes and lawful redescription. No distance, open-set relation, differentiable atlas, measure, or causal order appears in the definition. Such structures may be reconstructed later only if additional stability constraints force them. $\square$

6.4 Interior is not substance

The theorem also does not identify the persistence interior with substance or material stuff. It identifies the minimal non-vacuous role structure required for persistence to remain persistence. A later physical interpretation may use this role structure as the floor for concrete-domain participation, but the theorem itself is not a substance theorem.

7 Boundary and role fixation of the induced interior

The induced persistence interior must participate in the domain's role discipline. Otherwise it would either become a hidden object or dissolve into skin.

Definition 7.1 (Interior role slot). An interior role slot Slot_D is the admissibility-bearing role occupied by $[\text{Int}_D]_{\equiv_{\text{Int}}}$ when it supports standing retention, failure discrimination, transport, quotient identity, or reconstruction.

Definition 7.2 (Interior role-anchor equivalence). An interior role-anchor equivalence class $[R_D^{\text{Int}}]_{\equiv_{\text{Role}}}$ fixes the role of the persistence interior without selecting a representative of the interior.

Lemma 7.3 (Interior participation requires role anchoring). *If $[\text{Int}_D]_{\equiv_{\text{Int}}}$ participates in standing-bearing continuation, then it occupies a role slot fixed up to role-anchor equivalence.*

Proof. Participation in standing-bearing continuation is not mere presentation. It affects standing, failure, transport, or quotient behavior. By ATS role exhaustion, such effect is anchor or tensor load. If internal structure does not select a representative, boundary-trace fixation supplies role-anchor equivalence rather than representative selection. $\square$

Proposition 7.4 (Boundary discipline for the persistence interior). *The induced persistence interior is neither a hidden object nor a free representative. It is a role-fixed equivalence class constrained by boundary-trace and quotient discipline.*

Proof. By Minimal Persistence Interior, the interior is an equivalence class of differentiation structures. By the previous lemma, its participation requires role anchoring. Boundary-trace discipline fixes role participation without representative selection. Hence the interior is constrained enough to carry persistence but not over-specified as an object. $\square$

Corollary 7.5 (No hidden target object). *The non-vacuity theorem does not smuggle in a primitive object of the target regime. Any object-like interpretation would require later representative selection or additional reconstruction.*

8 Finite toy model: null collapse in a persistence graph

A toy model cannot prove the general theorem, but it can expose the exact failure mode of null collapse. The model uses a finite directed graph, a reconstruction map, an edge quotient, and a null-collapse quotient square.

Example 8.1 (Two-mark persistence graph). Let $G = (V, E)$ be a directed graph with vertices $V = \{a, b\}$. Let $m : V \rightarrow \{A, B\}$ be a mark map with $m(a) = A$ and $m(b) = B$. Let the

admissibility target be preservation of mark A under one-step continuation. The standing-positive edge is $e_{aa} : a \rightarrow a$. The standing-failing replacement edge is $e_{ab} : a \rightarrow b$. Thus

$$\text{Stand}(e_{aa}) = 1, \quad \text{Stand}(e_{ab}) = 0.$$

Define a reconstruction map ρ by

$$\rho(e_{xy}) = (m(x), m(y), \text{Stand}(e_{xy})).$$

Then $\rho(e_{aa}) = (A, A, 1)$ and $\rho(e_{ab}) = (A, B, 0)$, so the two histories occupy distinct reconstruction classes.

Definition 8.2 (Toy reconstruction equivalence). On the one-edge history fragment $H_G = \{e_{aa}, e_{ab}\}$, define

$$h_1 \equiv_\rho h_2 \quad :\Longleftrightarrow \quad \rho(h_1) = \rho(h_2).$$

Then $e_{aa} \not\equiv_\rho e_{ab}$. The quotient $H_G / \equiv_\rho$ therefore has two reconstruction classes: a standing-retaining class and a standing-failing replacement class. This is the finite instance of $\mathcal{R}_D$ used in the general theorem.

Definition 8.3 (Null-collapse quotient). The null-collapse quotient $q : V \rightarrow \{*\}$ sends both vertices to a single null vertex. It induces an edge quotient $q_E : E \rightarrow E_q$ by $q_E(e_{xy}) = * \rightarrow *$, and a collapsed reconstruction $\rho_q(e_{xy}) = (*, *, \sigma_q)$, where σ_q is any standing status retained after quotient.

The attempted null collapse can be displayed as the quotient square

$$\begin{array}{ccc} \{e_{aa}, e_{ab}\} & \xrightarrow{\rho} & \{(A, A, 1), (A, B, 0)\} \\ \downarrow q_E & & \downarrow \pi_q \\ \{* \rightarrow *\} & \xrightarrow{\rho_q} & \{(*, *, \sigma_q)\} \end{array} \quad (\text{TQ})$$

where π_q erases the mark distinction. If σ_q is constant, the standing-positive and standing-failing histories become reconstruction-equivalent. If σ_q is not constant, the quotient square commutes only by retaining a memory of the erased edge distinction, which is non-null structure.

Proposition 8.4 (Toy null-collapse failure). *Under the null-collapse quotient, the graph cannot preserve the distinction between e_{aa} and e_{ab} without importing extra non-null structure.*

Proof. The quotient sends both endpoints of both edges to $*$, so both edges become the same loop $* \rightarrow *$. If σ_q assigns the same standing status to both, then standing retention and failure collapse. If σ_q assigns different statuses to the two pre-quotient edges, then σ_q depends on the erased distinction between a and b , on a memory of the original edge, or on an access relation to a hidden reservoir. That rule is a non-null selector or trace memory. Therefore the null quotient either destroys the persistence distinction or imports non-null structure. $\square$

Corollary 8.5 (Finite model lesson). *Null collapse does not give a smaller realization of the same persistence target. It either erases the target distinction or preserves it by hidden non-null memory.*

Proposition 8.6 (Toy theorem-instance). *The two-mark graph instantiates the full theorem architecture. The original graph realizes NonVacRec because $H_G / \equiv_\rho$ has at least two standing-relevant classes. The constant null quotient realizes reconstruction collapse and standing failure. The nonconstant null quotient preserves standing only by hidden memory, selector load, access relation, or target change.*

Proof. The reconstruction classes before quotient are separated by $\rho(e_{aa}) \neq \rho(e_{ab})$, so the toy target has non-vacuous reconstruction. Applying q_E erases the edge distinction. If σ_q is constant, then e_{aa} and e_{ab} become equivalent and the target can no longer distinguish retained standing from failure; this is the finite analogue of reconstruction-null collapse. If σ_q is nonconstant, the output remembers a distinction erased by the quotient. That memory is an added selector, witness, access relation, or role anchor. If one instead changes the target so that both edges count the same, the original target has been abandoned. Thus the toy model realizes the same alternatives later formalized in the Exhaustion Partition.

The model also exhibits the failure of non-admissible reconstruction maps. Define a fake reconstruction $\tau : E \rightarrow \{0, 1\}$ after null collapse by stipulating $\tau(e_{aa}) = 1$ and $\tau(e_{ab}) = 0$. Since $q_E(e_{aa}) = q_E(e_{ab})$, τ is not quotient-invariant and cannot be induced from the collapsed target. It violates **R1**, **R3**, and **R6**: the distinction is not licensed by the collapsed fixed target, is not borne by licensed continuation after quotienting, and fails quotient discipline. If τ is nevertheless retained, it is an added selector or memory, so the candidate has moved to the imported-governance cell of the exhaustion partition. If τ is disallowed, the quotient degenerates to a single reconstruction class and realized persistence is lost. $\square$

9 Reconstruction stability and concrete-domain pressure

The persistence interior must not only exist; it must be stable enough to support reconstruction across lawful same-domain continuation. This stability is the first pressure toward concrete-domain participation, though not yet toward a particular physics model.

Definition 9.1 (Reconstruction stability). A persistence interior $[\text{Int}_D]_{\equiv_{\text{Int}}}$ is reconstruction-stable when lawful same-domain continuation and redescription preserve the standing-relevant reconstruction contrasts that define it, while standing failure remains fail-closed.

Lemma 9.2 (Persistence requires reconstruction). *If RealPers_D , the induced persistence interior is reconstruction-stable.*

Proof. If lawful continuation or redescription could erase the standing-relevant contrast while still counting as the same persistence, then persistence would be indistinguishable from failure or replacement. Fail-closed standing blocks this. Therefore the induced interior must be stable under the continuations licensed by D . $\square$

Lemma 9.3 (Trace-only reconstruction failure). *A trace, label, inscription, semantic reference, information record, or syntactic mark that does not preserve standing-relevant reconstruction contrasts cannot serve as the persistence interior.*

Proof. Such items are skin unless they carry anchor or tensor load. Skin has no independent gate authority. If the item carries anchor or tensor load, then it is not trace-only and supplies non-null structure. $\square$

Proposition 9.4 (Reconstruction Stability).

$$\text{MinReal}_D \wedge \text{RealPers}_D \implies [\text{Int}_D]_{\equiv_{\text{Int}}} \neq \emptyset \text{ and is reconstruction-stable.}$$

Proof. Nonemptiness follows from the Minimal Persistence Interior theorem. Stability follows from the persistence-requires-reconstruction lemma. $\square$

Proposition 9.5 (Concrete Reconstruction Necessity). *If $\text{MinReal}_D \wedge \text{RealPers}_D$, then the non-vacuous persistence interior is internal to the target regime in the following minimal sense: there exists a standing-relevant reconstruction contrast whose preservation or collapse changes persistence-bearing consequence inside D .*

Proof. By Reconstruction Non-Vacuity and Minimal Persistence Interior, D has a nonempty equivalence class of standing-relevant reconstruction contrasts. If every such contrast were external description, pure syntax, semantic reference, observer information, or presentation, then ATS role exhaustion would classify it as skin, and it could not alter standing retention, failure behavior, transport role, quotient identity, role participation, or lawful redescription. That would contradict realized persistence. Therefore at least one contrast is borne internally by the target regime as persistence-bearing reconstruction structure. This is the minimal sense in which the induced interior is concrete; it is not yet matter, spacetime, field content, objecthood, or a chosen representative. $\square$

Corollary 9.6 (Minimal concreteness). *Such internally reconstructible continuation structure constitutes the minimal sense in which a realized persistence regime is concrete. It is concrete because it is borne by the target regime as continuation-active role structure; it is minimal because it does not yet select a geometry, field, particle, material substrate, observer, or representative.*

Proof. External description and skin do not carry standing-bearing consequence. The induced interior does carry the contrasts without which realized persistence fails. This gives minimal target-internal concreteness, not a full physical ontology. $\square$

9.1 Concrete-domain pressure without concrete ontology

A reconstruction-stable persistence interior is not yet matter, spacetime, field, particle content, information content, or objecthood. But it is not mere abstract bookkeeping either: its contrasts are continuation-active and preserved by $\mathcal{F}_D$. This creates concrete-domain pressure in the minimal sense just defined. If a target is additionally required to be physical, the interior cannot be discharged into external description.

Definition 9.7 (Non-vacuous target interior under physical typing). NonVacInt_D holds when $[\text{Int}_D]_{\equiv \text{Int}} \neq \emptyset$, the interior is reconstruction-stable, and the target is additionally typed as physical in the limited sense of non-dischargeable domain participation.

Corollary 9.8 (Non-vacuous Interior Induction).

$$\text{PhysReal}_D \implies \text{NonVacInt}_D.$$

Proof. By definition, PhysReal_D includes realized persistence interpreted as non-dischargeable target participation. By Reconstruction Stability and Concrete Reconstruction Necessity, realized persistence induces a nonempty reconstruction-stable persistence interior borne internally by the target regime in the minimal role-theoretic sense. Since the target is physically typed in this limited sense, that interior is a non-vacuous target interior under physical typing. $\square$

Corollary 9.9 (No physically null realization).

$$\text{PhysReal}_D \implies \neg \text{PhysNull}_D.$$

Proof. PhysNull_D denies the continuation-sensitive reconstruction differentiation and non-skin role content required by NonVacInt_D . The previous theorem gives NonVacInt_D . Contradiction. $\square$

Theorem 9.10 (Minimal Realized Persistence Structure). *If $\text{MinReal}_D \wedge \text{RealPers}_D$, then the only structure forced by this paper is a nonempty, reconstruction-stable, target-internal minimal persistence interior:*

$$\text{NonVacRec}_D, \quad [\text{Int}_D]_{\equiv_{\text{int}}} \neq \emptyset, \quad \text{reconstruction-stability.}$$

No metric, topology, manifold, causal order, finite propagation law, field carrier, particle identity, material substrate, probability space, information primitive, observer, or representative selector follows at this stage.

Proof. Reconstruction Non-Vacuity gives NonVacRec_D . Minimal Persistence Interior gives $[\text{Int}_D]_{\equiv_{\text{int}}} \neq \emptyset$. Reconstruction Stability gives stability, and Concrete Reconstruction Necessity gives target-internal continuation-active role structure. Each theorem is formulated only in terms of admissibility, standing, reconstruction classes, lawful continuation, quotient discipline, and fail-closed standing. None introduces metricity, topology, fields, particles, material substrate, probability, observer information, or representative selection. Therefore the forced structure is exactly minimal realized persistence structure and nothing stronger. $\square$

10 Realization exhaustion

The last formal step is to make the escape-route exhaustion explicit. A reader should not have to infer why the listed routes are the only routes.

Definition 10.1 (Null-realization escape route). A null-realization escape route is a same-domain strategy that attempts to satisfy RealPers_D or PhysReal_D while denying NonVacRec_D , $[\text{Int}_D]_{\equiv_{\text{int}}} \neq \emptyset$, or NonVacInt_D .

Theorem 10.2 (Exhaustion Partition). *Let X be any candidate same-domain regime offered as realized persistence. Exactly one of the following partition cells applies with respect to the original fixed target after normalization by target, access, and role import:*

- P1.** *X preserves standing-relevant continuation-sensitive reconstruction distinctions.*
- P2.** *X loses those distinctions while retaining only skin, trace, syntax, or external semantics.*
- P3.** *X loses those distinctions and exits minimal realization, becoming AMetric , inert, or unevaluated.*
- P4.** *X appeals to a latent reservoir whose distinctions are inaccessible to every $r \in \mathcal{F}_D$.*
- P5.** *X recovers the distinctions by adding a selector, witness, gate, quotient, transport rule, role anchor, access relation, or memory.*
- P6.** *X changes the admissibility gate, evaluated fragment, failure semantics, target vocabulary, or comparison relation.*

Only P1 remains a same-domain realized persistence candidate, and P1 is non-vacuous.

Proof. Either standing-relevant reconstruction distinctions are preserved or they are not. If preserved, **P1** holds and non-vacuity follows by definition. If not preserved, then the remaining claim to realization must rest on skin-like remainder, no additional role, inaccessible latent structure,

imported accessible role structure, or altered target conditions. Skin-like remainder gives **P2**. No additional role gives AMetric, inert, or unevaluated status, hence **P3**. Inaccessible latent structure gives **P4**, but by Hidden Reconstruction Reservoir Exclusion it cannot function as realized persistence structure for D . Imported accessible role structure gives **P5**. Altered target conditions give **P6**. These cells are mutually exclusive after normalizing by whether the original fixed target is retained, whether alleged structure is accessible to $\mathcal{F}_D$, and whether non-skin role structure is imported. No additional persistence-bearing reconstruction category exists outside the partition, because any proposed category either preserves admissible continuation-sensitive distinction, fails realization-sensitive reconstruction, imports accessible non-null role structure, remains inaccessible, or changes the target. The only cell compatible with same-domain realized persistence is **P1**, which is non-vacuous. $\square$

Proposition 10.3 (Route Collapse). *Every null-realization escape route collapses into AMetric inertness, skin-only description, inaccessible hidden reservoir, hidden non-null governance, standing failure, or scope change.*

Proof. A null-realization escape route denies **P1**. By the Exhaustion Partition, it falls into **P2**, **P3**, **P4**, **P5**, or **P6**. These are respectively skin-only description, AMetric/inert exit, inaccessible hidden reservoir, imported non-null governance, and scope change. By Hidden Reconstruction Reservoir Exclusion, **P4** cannot carry realized persistence for the fixed domain. If the route also claims to preserve standing while losing the distinction between retention and failure, it additionally collapses to standing failure by fail-closed standing. Hence no route preserves same-domain null realization. $\square$

Corollary 10.4 (No physically null realization, exhaustion form). *No same-domain candidate satisfies $\text{PhysReal}_D \wedge \text{PhysNull}_D$.*

Proof. If PhysReal_D , the candidate must fall into **P1** of the Exhaustion Partition. But **P1** is non-vacuous. Therefore PhysNull_D is false. $\square$

11 Scope of subsequent reconstructions

The result proved here is intentionally minimal. It does not claim unique reconstruction of full physical law, dynamical structure, geometry, quantum theory, field content, constants, thermodynamics, vacuum physics, or cosmology. It establishes only that realized persistence in a fixed admissible continuation regime cannot be reconstruction-null.

Consequently, any later reconstruction of locality, neighborhood structure, metricity, finite propagation, field-like carriers, particle identity, gauge structure, mass, vacuum structure, or temporal ordering must be derived from additional persistence, repair, transport, perturbation, and boundary-trace constraints. None of those structures is used as a premise in the present argument.

12 Relation to existing physical and reconstruction frameworks

This section positions the result relative to nearby reconstruction programs. It is not a literature survey. Operational reconstructions of quantum theory often begin from preparations, transformations, measurement outcomes, probabilities, information constraints, or composition rules [18, 19, 20, 21]. Algebraic approaches to field theory encode physical content in algebras, nets,

covariance, locality, and state spaces [16, 17]. Structural-realist accounts emphasize structure, relations, or invariance as ontologically or epistemically privileged [13, 14, 15]. The present theorem begins at a different level. It asks what must hold of a same-domain target that already carries realized persistence, before measurement, probability, metricity, locality, field structure, or object ontology has been introduced.

12.1 Operational and information-theoretic reconstructions

The reconstruction relation used here is not observer knowledge, information flow, entropy, coding, communication, or measurement accessibility. A persistence-sensitive reconstruction map is licensed by admissibility-bearing continuation inside the fixed domain. Information-theoretic structures may later become admitted carriers or descriptions, but they are not the source of the non-vacuity theorem. The result is therefore pre-operational rather than a competing axiom system for quantum theory.

12.2 Algebraic and categorical presentation

The argument is compatible with algebraic or categorical presentation: fixed domains, histories, reconstruction maps, quotients, and role-preserving transformations can be organized in those idioms. No algebraic net, locality axiom, categorical universal property, or field algebra is required for the theorem. The proof depends only on fixed-domain admissibility, constrained reconstruction maps, quotient discipline, and fail-closed standing.

12.3 Structural realism and ontology

The result is compatible with structural readings in the limited sense that it privileges role, invariance, and reconstruction over primitive objecthood. It does not rest on a philosophical thesis of structural realism. The proof uses fixed-domain admissibility, reconstruction equivalence, fail-closed standing, and boundary discipline. Those constraints are sufficient for the non-vacuity result.

12.4 Metric, field, particle, and vacuum interpretations

The theorem neither assumes nor denies metrics, fields, particles, or vacuum structures inside later admitted regimes. It denies only that those structures are available as premises of the first non-vacuity result. If metric structure, field carriers, particle classes, or vacuum structure are reconstructed later, they must arise from additional stability and continuation constraints. The present theorem implies only that a realized physical vacuum, if typed as a persistence-bearing target, is not total nullity.

12.5 Cosmology, modality, and ensembles

The theorem does not derive a cosmological origin, a temporal beginning, a universe from nothing, or a primitive modal ontology. It does not quantify over possible-world ensembles, probability measures over worlds, or anthropic conditioning. Such structures may be introduced only as admitted structures inside a fixed target or as explicit scope changes. They cannot supply an AMetric default for realized persistence.

13 Conclusion

The analysis establishes the first result in the realized-physics ladder. Within a fixed admissible continuation regime, realized persistence minimally requires non-trivial continuation-sensitive reconstruction structure internal to the regime itself. Null realization, inaccessible latent reservoirs, arbitrary reconstruction maps, and reconstruction-degenerate continuations fail to preserve standing-bearing persistence under lawful same-domain continuation.

The theorem-level result is:

$$\text{MinReal}_D \wedge \text{RealPers}_D \Rightarrow \text{NonVacRec}_D \Rightarrow [\text{Int}_D]_{\equiv_{\text{Int}}} \neq \emptyset.$$

The physics-facing corollary is:

$$\text{PhysReal}_D \Rightarrow \neg \text{PhysNull}_D.$$

The physical interpretation is positive rather than merely defensive: Paper 1 derives the minimal non-null role of physical realization. It does not yet derive locality, metricity, propagation, field carriers, particles, constants, thermodynamics, vacuum physics, or cosmology. It gives those later analyses their required floor: physical realization, if it carries persistence at all, must be non-vacuous in standing-relevant continuation structure.

A Normal form of the result

For later reference, the theorem can be recorded in normal form.

Proposition A.1 (Normal Form). *Let D be a fixed domain. Assume:*

- A1.** *D has a stable admissibility gate on an evaluated fragment.*
- A2.** *D has lawful same-domain continuation.*
- A3.** *D has a persistence-sensitive reconstruction interface with reconstruction maps satisfying **R1–R7**.*
- A4.** *D treats standing failure fail-closedly.*
- A5.** *D carries realized standing-bearing persistence.*

Then:

- C1.** *D has non-identical continuation-sensitive reconstruction classes.*
- C2.** *D has nonempty standing-relevant differentiation Diff_D .*
- C3.** *D has a nonempty minimal persistence interior $[\text{Int}_D]_{\equiv_{\text{Int}}}$.*
- C4.** *That interior is internal to the target regime in the minimal sense of Concrete Reconstruction Necessity.*
- C5.** *The forced structure is only minimal realized persistence structure; stronger physics-facing structures are deferred.*
- C6.** *If the target is physical realization, D is not physically null.*

Proof. **A1–A4** are MinReal_D . With **A5**, Reconstruction Non-Vacuity gives **C1**. Persistence Differentiation gives **C2**. Minimal Persistence Interior gives **C3**. Concrete Reconstruction Necessity gives **C4**. Minimal Realized Persistence Structure gives **C5**. Non-vacuous Interior Induction and its corollary give **C6** under physical target typing. $\square$

B Notation index

Symbol	Meaning
D	Fixed domain.
E_D	Evaluated fragment of D .
Adm_D	Admissibility gate on D .
Stand_D	Standing predicate; reuse-stable admissibility.
PersCap_D	Persistence-capability: the target question can be asked.
MinReal_D	Minimal realization: stable admissible continuation capable of persistence-sensitive reconstruction.
RealPers_D	Realized persistence regime is in force.

Symbol	Meaning
PhysReal_D	Realized physical target: persistence as non-dischargeable domain participation.
PhysNull_D	Purported physically null same-domain realization.
Hist_D	Same-domain continuation histories.
Rec_D	Reconstruction interface for standing-bearing consequence.
$\mathcal{F}_D$	Admissibility-preserving persistence-sensitive reconstruction maps satisfying R1–R7 .
$\equiv_{\text{Rec}}$	Reconstruction equivalence.
$\mathcal{R}_D$	Continuation-sensitive reconstruction classes.
NonVacRec_D	Non-vacuous reconstruction: non-identical standing-relevant reconstruction classes.
Diff_D	Standing-relevant interior differentiations.
$\equiv_{\text{Int}}$	Persistence-interior equivalence under lawful redescription.
$[\text{Int}_D]_{\equiv_{\text{Int}}}$	Minimal persistence interior equivalence class.
NonVacInt_D	Non-vacuous target interior under physical target typing.
$[G_D]_{\equiv_{\text{Gov}}}$	Governance equivalence class.
$[L_D]_{\equiv_{\text{Loc}}}$	Continuation-locus equivalence class.
$[R_D]_{\equiv_{\text{Role}}}$	Role-anchor equivalence class.
$[P_D]_{\equiv_{\text{Pers}}}$	Persistence-regime equivalence class.

C Route-collapse normal forms

Route	Attempted role	Collapse reason
Mute nullity	Avoid all interior content.	Not evaluated; if used to govern, it is no longer mute.
AMetric default	Let nullity win by absence of alternatives.	Default selection is a selector; selectors are not available at the AMetric boundary without admitted structure.
Empty extension	Admit no occupants but keep the target.	May be governed, but without standing-retained continuation it is not realized persistence.
Reconstruction-null persistence	Collapse all histories to one class.	Cannot distinguish standing retention from failure or lawful continuation from replacement.
Trace-only continuation	Preserve order while erasing interior.	Order alone does not preserve identity, transport, quotient, role, or failure behavior.
Formal-only description	Treat proof or syntax as realization.	Inscription may describe a target but does not carry standing-bearing domain participation.

Route	Attempted role	Collapse reason
Semantic-only reference	Let meaning or denotation supply target interior.	Reference without transportable consequence retention is not realized persistence.
Skin-only presentation	Retain notation, labels, or coordinates only.	Skin has no gate authority; any authority is anchor or tensor load.
Information-only record	Treat observer information or a code as the carrier.	Information with no admissibility-bearing role is skin; if it carries role load, it is non-null structure.
Global standing	Let standing be everywhere.	If no distinguishable persistence role exists, reconstruction fails; if roles exist, the regime is non-null.
Diffuse standing	Let standing be nowhere in particular.	Diffusion without reconstructible persistence roles destroys persistence identity.
Hidden reservoir	Preserve erased distinctions in an inaccessible latent layer.	If inaccessible to every persistence-sensitive reconstruction map, it is realization-inert; if accessible, the access relation is non-null.
Hidden selector	Quietly pick the interior or representative.	The selector is non-null governance and must be admitted.
Same-act repair	Allow null failure to become standing-positive later.	Violates fail-closed standing; legitimate improvement is a new act or scope.
Lower generator	Generate realization beneath admissibility.	Lower layer is either AMetric inert or crosses the boundary by non-null structure.
Scope change	Modify target, domain, gate, or role vocabulary.	Not a same-domain null realization of the original target.

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